A2 · Unit 5 · WPH15 Thermodynamics, Radiation, Oscillations & Cosmology

Oscillations

Pendulums, guitar strings, car suspensions, the atoms in a crystal and the swaying of a skyscraper all oscillate. Most of them follow the same simple rule, simple harmonic motion, and once you understand that one rule you can predict their periods, their speeds and their energies. Then we look at what happens when oscillations are damped, and when something pushes them at just the right frequency.

What makes motion simple harmonic?

Every oscillator has an **equilibrium position** where it would happily sit still. Pull it away and a **restoring force** pushes it back. It arrives back at equilibrium moving fast, overshoots because of its inertia, slows down on the other side, and comes back again.

The motion is **simple harmonic** when the restoring force is **proportional to the displacement**: pull twice as far, get twice the force back. That is exactly what a spring obeying Hooke's law does.

The condition for simple harmonic motion

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$$F = -kx$$

F: restoring force / N k: constant (the stiffness) / N m^{-1} x: displacement from equilibrium / m

The **minus sign** is the important part: the force always points the opposite way to the displacement, back towards equilibrium. Since $F = ma$, the acceleration behaves in the same way: it is proportional to the displacement and always directed towards the equilibrium position.

DEFINITION

Simple harmonic motion

Simple harmonic motion (SHM) is an oscillation in which the acceleration of the object is **directly proportional to its displacement** from the equilibrium position and is **always directed towards** the equilibrium position.

Spotting SHM

To decide whether something moves with SHM, work out the resultant force on it at a displacement x. If it comes out as $F = -(\text{a constant}) \times x$, the motion is simple harmonic.

System	SHM?	Why
mass on a spring	yes	Hooke's law gives $F = -kx$, as long as the spring stays within its limit of proportionality
simple pendulum	yes, for small swings	the restoring force is $-mg\sin\theta$, which is close to proportional to x only when θ is small
a floating block pushed down and released	yes	the extra upthrust is proportional to how far it is pushed down
a ball bouncing on the floor	no	while it is in the air the force on it is mg, the same at every height, not proportional to its displacement

The equations of SHM

Combining $F = -kx$ with $F = ma$ gives $a = -(k/m)x$. The constant k/m is always positive, so we write it as ω^2, and ω turns out to be the **angular frequency** of the oscillation.

Defining equation of S

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$$a = -\omega^2 x$$

a: acceleration / m s^{-2}

ω: angular frequency / rad s^{-1}

Period, frequency and

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$$T = \frac{1}{f} = \frac{2\pi}{\omega}, \qquad \omega = 2\pi f$$

T: period / s f: frequency / Hz

If you let go of the oscillator at $x = A$ (its **amplitude**, the maximum displacement) at $t = 0$, its displacement, velocity and acceleration at any time are:

Displacement

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$$x = A\cos\omega t$$

Velocity

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$$v = -A\omega\sin\omega t$$

Acceleration FORMULA SHEET

$$a = -A\omega^2 \cos \omega t$$

Compare the first and last: $a = -\omega^2(A \cos \omega t) = -\omega^2 x$, the defining equation again. The biggest speed is $v_{\max} = A\omega$, reached as the object passes through equilibrium. The biggest acceleration is $a_{\max} = A\omega^2$, at the two ends of the motion.

Where the cosine and the ω come from

Picture a point moving steadily round a circle of radius A with angular velocity ω (as in circular motion). After time t it has turned through an angle $\theta = \omega t$. Shine a light so that its shadow falls on a diameter: the shadow moves back and forth with displacement $A \cos \omega t$, which is SHM. The point's speed round the circle is $A\omega$, and its centripetal acceleration $A\omega^2$ points to the centre. Take the components of these along the diameter and you get exactly $v = -A\omega \sin \omega t$ and $a = -A\omega^2 \cos \omega t$. So SHM is “circular motion seen side-on”, and ω is the angular velocity of that imaginary point.

EXAM TIP

Put your calculator in **radians** for $\cos \omega t$ and $\sin \omega t$: ωt is an angle in radians. Also, $x = A \cos \omega t$ only applies if timing starts at maximum displacement. If timing starts as the object passes through equilibrium, the displacement is $x = A \sin \omega t$ instead.

Displacement, velocity and acceleration graphs

The three graphs are linked by gradients, just as in the Motion topic:

- the **gradient of the displacement–time graph is the velocity**. Where x is at a peak or a trough the graph is flat, so $v = 0$. Where x crosses zero the graph is steepest, so the speed is greatest;
- the **gradient of the velocity–time graph is the acceleration**. So a is zero where v is at its biggest, and largest where v is zero.

All three are sine or cosine curves with the same period. The velocity is a quarter of a cycle out of step with the displacement, and the acceleration is exactly half a cycle out of step: it is the displacement graph turned upside down and stretched by ω^2, because $a = -\omega^2 x$.

Position	Displacement	Velocity	Acceleration

at either end	$\pm A$ (maximum)	zero	maximum, $\mp A\omega^2$, pointing to the middle
passing through equilibrium	zero	maximum, $\pm A\omega$	zero

Mass–spring systems and pendulums

For a particular oscillator, ω comes from comparing its force equation with $a = -\omega^2 x$. That gives the two standard results.

Period of a mass on a spring

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A mass m on a spring of stiffness k feels a restoring force $F = -kx$. Newton's second law, $F = ma$, gives

$$ma = -kx \quad \Rightarrow \quad a = -\frac{k}{m}\,x$$

2. This has the form $a = -\omega^2 x$, so the motion is SHM with

$$\omega^2 = \frac{k}{m} \quad \Rightarrow \quad \omega = \sqrt{\frac{k}{m}}$$

3. Use $T = 2\pi/\omega$:

$$\boxed{T = 2\pi\sqrt{\frac{m}{k}}}$$

A heavier mass has more inertia, so it oscillates more slowly. A stiffer spring pulls back harder, so it oscillates faster. The amplitude doesn't appear at all.

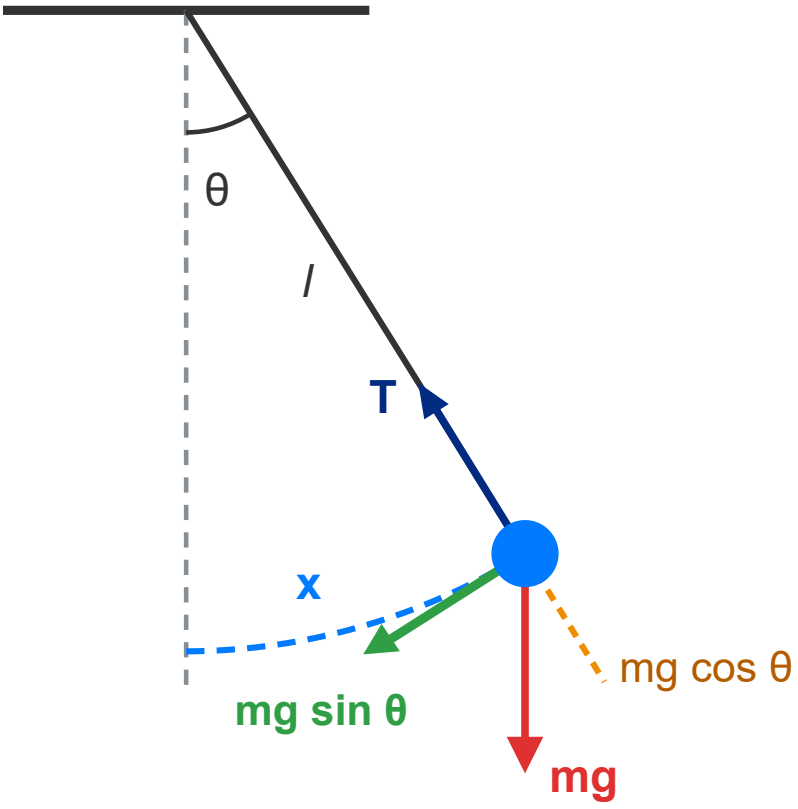
Mass–spring oscillator

FORMULA SHEET

$$T = 2\pi\sqrt{\frac{m}{k}}$$

m: oscillating mass / kg k: spring constant / N m^{-1}

The same formula works for a spring hanging vertically. Gravity just moves the equilibrium position down; the extra force for each extra bit of stretch is still kx.



The component of the weight along the arc, $mg\sin\theta$, pulls the bob back towards equilibrium. The tension and $mg\cos\theta$ act along the string and don't change its speed along the arc.

Period of a simple pendulum (small angles)

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A bob of mass m on a string of length l is pulled aside by an angle θ. The component of its weight along the arc pulls it back towards equilibrium:

$$F = -mg\sin\theta$$

2. For **small angles** (in radians), $\sin\theta \approx \theta$. The bob's displacement along the arc is $x = l\theta$, so $\theta = x/l$:

$$F \approx -mg\theta = -\frac{mg}{l}\,x$$

At 10°, $\sin\theta$ and θ differ by only 0.5%, so this works well for swings of up to about 10°.

3. Divide by m to get the acceleration. The mass cancels:

$$a = -\frac{g}{l}\,x \quad \Rightarrow \quad \omega^2 = \frac{g}{l}$$

4. Use $T = 2\pi/\omega$:

$$\boxed{T = 2\pi\sqrt{\frac{l}{g}}}$$

The period depends only on the length and on g: not on the mass of the bob, and not on the amplitude while the swing stays small.

Simple pendulum

FORMULA SHEET

$$T = 2\pi\sqrt{\frac{l}{g}}$$

l: length from pivot to centre of bob / m g: gravitational field strength / N kg^{-1}

EXAM TIP

When you time a pendulum, time 10 or 20 complete oscillations and divide, so the reaction-time error is shared out. Start and stop the timer as the bob passes the **centre**, where it moves fastest, using a marker behind it.

WORKED EXAMPLE

A 0.40 kg mass hangs on a spring with $k = 60\ \text{N m}^{-1}$. It is pulled down 5.0 cm and released at $t = 0$. Calculate (a) the period, (b) the maximum speed, (c) the maximum acceleration, and (d) the displacement and velocity at $t = 0.10\ \text{s}$.

Show solution

(a) $T = 2\pi\sqrt{m/k} = 2\pi\sqrt{0.40/60} = 0.513\ \text{s}$, and $\omega = \sqrt{k/m} = \sqrt{150} = 12.2\ \text{rad s}^{-1}$.

(b) $v_{\max} = A\omega = 0.050 \times 12.25 = 0.61\ \text{m s}^{-1}$

(c) $a_{\max} = A\omega^2 = 0.050 \times 150 = 7.5\ \text{m s}^{-2}$

(d) $\omega t = 12.25 \times 0.10 = 1.225\ \text{rad}$ (radians!), so

$$x = A\cos\omega t = 0.050\cos(1.225) = 0.017\ \text{m} = 1.7\ \text{cm}$$

$$v = -A\omega\sin\omega t = -0.612 \times \sin(1.225) = -0.58\ \text{m s}^{-1}$$

The minus sign means it is moving back towards (and about to pass through) the equilibrium position, which makes sense: 0.10 s is just under a quarter of a period.

WORKED EXAMPLE

A student times 20 oscillations of a pendulum of length 0.800 m and gets 35.9 s. Calculate a value for g.

Show solution

$T = 35.9/20 = 1.795\ \text{s}$. Squaring $T = 2\pi\sqrt{l/g}$ and rearranging:

$$g = \frac{4\pi^2 l}{T^2} = \frac{4\pi^2 \times 0.800}{1.795^2} = 9.80\ \text{N kg}^{-1}$$

A better method is to use several lengths and plot T^2 against l: the gradient is $4\pi^2/g$.

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Big swings take longer

The small-angle approximation hides something: for larger swings, $\sin\theta$ is noticeably smaller than θ, so the restoring force is weaker than SHM predicts and the pendulum takes longer to come back. The period grows roughly as $T \approx 2\pi\sqrt{l/g}\left(1 + \theta_0^2/16\right)$ with θ_0 in radians. At 30° the period is about 1.7% longer than the formula says, and at 90° about 18% longer. Galileo thought the pendulum's period was exactly independent of amplitude, and clockmakers later had to keep the swing small (or use special shapes of pendulum) to keep good time.

Energy in SHM

As an oscillator moves, energy keeps changing form. For a mass on a spring it swaps between **kinetic energy** of the mass and **elastic potential energy** stored in the spring (for a pendulum, gravitational potential energy). If nothing dissipates energy, the total stays constant: the oscillation is **undamped**.

- At the ends ($x = \pm A$) the mass is momentarily at rest: all the energy is potential.
- At equilibrium ($x = 0$) the spring has its natural stretch: all the energy is kinetic, and the speed is $v_{\max}$.

Total energy of a mass–spring oscillator

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. The elastic potential energy stored in a spring is $\frac{1}{2}F\Delta x$. With $F = kx$, at displacement x it stores

$$E_{\mathrm{p}} = \tfrac{1}{2}kx^2$$

2. At the end of the swing, $x = A$ and the mass is at rest, so all the energy is potential:

$$\boxed{E_{\mathrm{total}} = \tfrac{1}{2}kA^2}$$

3. Check at equilibrium, where all of it is kinetic. Using $v_{\max} = A\omega$ and $\omega^2 = k/m$:

$$\tfrac{1}{2}mv_{\max}^2 = \tfrac{1}{2}mA^2\omega^2 = \tfrac{1}{2}mA^2\frac{k}{m} = \tfrac{1}{2}kA^2$$

The two agree, as energy conservation says they must. The energy is proportional to A^2: double the amplitude and the oscillator holds four times as much energy.

At any point in between, $E_k + E_p = \frac{1}{2}kA^2$. Because both depend on squares, each one goes through two maxima per oscillation: kinetic and potential energy oscillate at **twice** the frequency of the motion.

WORKED EXAMPLE

For the 0.40 kg mass on the 60 N m⁻¹ spring with amplitude 5.0 cm, find (a) the total energy and (b) the speed of the mass when its displacement is 3.0 cm.

Show solution

(a) $E = \frac{1}{2}kA^2 = 0.5 \times 60 \times 0.050^2 = 0.075$ J

(b) At $x = 0.030$ m: $E_p = 0.5 \times 60 \times 0.030^2 = 0.027$ J, so $E_k = 0.075 - 0.027 = 0.048$ J.

$$v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 0.048}{0.40}} = 0.49 \text{ m s}^{-1}$$

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A shortcut: v = ±ω√(A² − x²)

The energy method above can be turned into a general formula for the speed at any displacement.

Speed at displacement x

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1. Energy conservation at displacement x:

$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$$

2. Multiply by $2/m$ and use $k/m = \omega^2$:

$$v^2 = \omega^2\left(A^2 - x^2\right) \quad \Rightarrow \quad v = \pm\omega\sqrt{A^2 - x^2}$$

Check with the worked example: $12.25 \times \sqrt{0.050^2 - 0.030^2} = 12.25 \times 0.040 = 0.49$ m s^{-1}. *The ± is because the mass passes each point going both ways.*

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Why x = A cos ωt solves a = −ω²x

If you know calculus, you can check the formula-sheet equations directly. Acceleration is the second derivative of displacement.

Differentiating x = A cos ωt twice

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1. Differentiate once to get the velocity:

$$v = \frac{dx}{dt} = \frac{d}{dt}\left(A\cos\omega t\right) = -A\omega\sin\omega t$$

2. Differentiate again to get the acceleration:

$$a = \frac{dv}{dt} = -A\omega^2\cos\omega t = -\omega^2\left(A\cos\omega t\right) = -\omega^2 x$$

So $x = A\cos\omega t$ satisfies the SHM equation. $A\sin\omega t$, or any mixture of the two, also works: that is how the starting conditions are fitted.

Damping

Real oscillations die away. Air resistance, friction and the slight heating of the spring as it flexes all do work against the motion, so energy is transferred out of the oscillator, mostly as thermal energy in the surroundings. Forces like this cause **damping**, and the amplitude gets smaller and smaller.

DEFINITION

Damping

Damping is the effect of **dissipative forces** (such as friction and air resistance) that transfer energy away from an oscillating system, so the amplitude of the oscillations decreases.

- **Light damping:** the amplitude shrinks gradually over many oscillations, and the period stays almost the same (a pendulum in air).
- **Heavy damping:** the system creeps slowly back to equilibrium without oscillating at all (a pendulum in thick oil).
- In between, **critical damping** returns the system to equilibrium in the shortest time without overshooting. Car suspensions are designed to be close to this, so the car doesn't keep bouncing after a bump.

Energy is still conserved in a damped oscillator. It is just shared with the surroundings: kinetic + potential energy of the oscillator + energy dissipated = the energy it started with, $\frac{1}{2}kA_0^2$. Each cycle the oscillator loses a fraction of its energy, which is why the amplitude keeps falling.

Plastic deformation as a damper

A ductile material such as steel, lead or copper can be stretched or bent permanently (plastic deformation). When that happens, the work done to deform it is **not given back** when the force is removed: it ends up as thermal energy inside the material. On a force–extension graph, the energy dissipated in each loading and unloading cycle is the area enclosed between the two curves.

Engineers use this to protect buildings in earthquake zones. Dampers made of ductile metal (and bearings with lead cores) are built into the structure so that they deform plastically as the building sways. Each sway takes energy out of the oscillation, so the amplitude falls quickly instead of building up to a level that would damage the main structure.

Free and forced oscillations: resonance

DEFINITION

Free and forced oscillations

A **free oscillation** happens when a system is displaced and then left alone, with no external periodic force. It oscillates at its own **natural frequency**, f_0.

A **forced oscillation** happens when a periodic driving force acts on the system. Once it has settled, the system oscillates at the **driving frequency**, whatever its natural frequency is.

Resonance occurs when the driving frequency equals the natural frequency of the system. The amplitude of the oscillations is then a maximum, because energy is transferred from the driver to the oscillator most efficiently.

Think of pushing a child on a swing. Push once per swing, at the right moment, and every push is in the same direction as the swing is moving. Each push does positive work, so energy builds up cycle after cycle. The amplitude only stops growing when the energy lost to damping in each cycle equals the energy the pushes add. Push at the wrong frequency and some pushes act against the motion, taking energy back out, so the swing never gets far.

How the amplitude of a forced oscillation depends on the driving frequency:

- **Well below** f_0: the system simply follows the driver, with roughly the driver's amplitude.
- **At** f_0: resonance, and the amplitude is at its peak.
- **Well above** f_0: the system can't keep up and the amplitude falls towards zero.

Damping changes the shape of this resonance curve. More damping makes the peak **lower** and **broader** (the system responds over a wider range of frequencies, but never very strongly), and the peak moves slightly **below**

f_0. Very light damping gives a tall, sharp peak.

Resonance in real life

Resonance is useful in musical instruments (the air in a guitar body or a flute resonates), in tuning a radio to one station, and in the quartz crystal that keeps a watch on time. It can also be destructive. A building whose natural frequency matches the shaking of an earthquake can sway with a huge amplitude, which is why tall buildings are fitted with dampers. In 2000 the Millennium Bridge in London began swaying sideways as soon as it opened, because the walkers fell into step with its natural frequency, and it had to be closed until dampers were fitted.

WORKED EXAMPLE

A tall building in an earthquake zone has a natural frequency of 0.8 Hz. Explain why it could be badly damaged by an earthquake with ground vibrations of about 0.8 Hz, and why engineers fit dampers.

Show solution

The ground shaking is a periodic driving force, so the building performs forced oscillations. When the driving frequency (0.8 Hz) equals the building's natural frequency, resonance occurs: energy is transferred to the building most efficiently, so the amplitude of its sway becomes very large and can exceed what the structure can withstand.

Dampers dissipate energy from the oscillations (for example by plastic deformation of ductile metal, or by friction in fluid dampers). This makes the resonance peak much lower, so the maximum amplitude is smaller, and it makes the oscillations die away quickly once the shaking stops.

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The Q factor: how sharp is the resonance?

Physicists describe the sharpness of a resonance with the quality factor, $Q = f_0/\Delta f$, where Δf is the width of the resonance curve at a certain height. A lightly damped system has a high Q: a tuning fork has Q of around 1000 and a quartz watch crystal tens of thousands, so each responds only to a very narrow range of frequencies. A car suspension has Q of about 1, so it barely resonates at all. A high-Q system also keeps ringing for many oscillations after it is struck, which is why a tuning fork sounds for so long.

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Marching soldiers and the Tacoma Narrows Bridge

Soldiers are ordered to break step when crossing a bridge. In 1831 the Broughton suspension bridge near Manchester collapsed under marching troops, and a regular, strong footfall near a bridge's natural frequency could drive it towards resonance. The famous film of the Tacoma Narrows Bridge twisting itself apart in 1940

is often used as an example of resonance, but it wasn't simple forced resonance: the wind was steady, not periodic. The bridge's own twisting changed how the wind pushed on it, feeding energy in on every cycle. This self-excited process is called aeroelastic flutter.

Core Practical 16: finding an unknown mass

CORE PRACTICAL

Determine the value of an unknown mass using the resonant frequencies of the oscillation of known masses

Hang a spring from a vibration generator connected to a signal generator, and attach a known mass m. Slowly increase the driving frequency and find the frequency at which the mass oscillates with the **largest amplitude**. That is resonance, so the driving frequency equals the natural frequency of the mass–spring system. Repeat for several known masses.

From $T = 2\pi\sqrt{m/k}$, squaring gives

$$T^2 = \frac{4\pi^2}{k}\, m$$

So with $T = 1/f$, a graph of T^2 **against** m is a straight line with gradient $4\pi^2/k$. Now hang the unknown mass, find its resonant frequency, work out T^2, and read its mass from the graph (or calculate $m = T^2/\text{gradient}$).

- Find each resonant frequency by approaching it from below and from above, and take the mean. Fine-tune the frequency in small steps near the peak.
- Keep the driving amplitude small, so the spring stays within its limit of proportionality and never goes slack.
- The spring has mass too, and part of it oscillates. This is a systematic error that gives the line a small intercept, but using the graph (rather than a single calculation) deals with it, because the unknown mass is read from the same line.
- **Safety:** clamp the stand firmly to the bench (or use a counterweight) so it can't topple, and keep your feet clear in case a mass falls.

WORKED EXAMPLE

A student's graph of T^2 against m has a gradient of 1.58 s² kg⁻¹. An unknown mass on the same spring resonates at 1.40 Hz. Calculate (a) the spring constant and (b) the unknown mass.

Show solution

(a) The gradient is $4\pi^2/k$, so

$$k = \frac{4\pi^2}{1.58} = 25.0 \text{ N m}^{-1}$$

(b) $T = 1/f = 1/1.40 = 0.714$ s, so $T^2 = 0.510$ s^2 and

$$m = \frac{T^2}{\text{gradient}} = \frac{0.510}{1.58} = 0.32 \text{ kg}$$

Quick checks

QUICK CHECK

Which equation describes an object moving with simple harmonic motion?

- $a = \omega^2 x$
- $a = -\omega^2 x$ ✓
- $a = -\omega x^2$
- $a = -g$

The acceleration must be proportional to the displacement (x, not x^2) and opposite to it, hence the minus sign. $a = -g$ is a constant acceleration, like a bouncing ball in the air, which is not SHM.

QUICK CHECK

An object in SHM passes through its equilibrium position. Which statement is correct?

- Its velocity is zero and its acceleration is a maximum.
- Both its velocity and its acceleration are a maximum.
- Its speed is a maximum and its acceleration is zero.
- Both its velocity and its acceleration are zero.

At $x = 0$, $a = -\omega^2 x = 0$: there is no restoring force there. All of the energy is kinetic, so the speed is at its greatest, $A\omega$.

QUICK CHECK

A pendulum clock is taken to the Moon, where g is one sixth of its value on Earth. By what factor does the period change?

- It becomes 6 times longer.
- It becomes about 2.4 times longer.
- It becomes about 2.4 times shorter.
- It doesn't change, because the period doesn't depend on mass.

$T = 2\pi\sqrt{l/g}$, so $T \propto 1/\sqrt{g}$. Dividing g by 6 multiplies T by $\sqrt{6} = 2.45$. The period doesn't depend on the bob's mass, but it does depend on g.

QUICK CHECK

The amplitude of a mass–spring oscillator is doubled. What happens to its period and to its total energy?

- The period doubles and the energy doubles.
- The period is unchanged and the energy doubles.
- The period is unchanged and the energy is four times bigger.
- The period doubles and the energy is four times bigger.

$T = 2\pi\sqrt{m/k}$ has no A in it. The total energy is $\frac{1}{2}kA^2$, so doubling A multiplies it by 4.

QUICK CHECK

The damping of a driven oscillator is increased. How does its resonance curve (amplitude against driving frequency) change?

- The peak becomes taller and narrower.
- The peak becomes lower and broader.
- The peak moves to a much higher frequency.
- The curve doesn't change, only how quickly it is reached.

More damping dissipates more energy per cycle, so the amplitude at resonance is smaller and the response is spread over a wider range of frequencies. The peak also moves slightly *below* the natural frequency, not above it.

QUICK CHECK

A mass on a spring has a period of 0.50 s. The mass is replaced by one four times as heavy. What is the new period?

- 0.25 s
- 0.50 s
- 1.0 s
- 2.0 s

$T \propto \sqrt{m}$, so four times the mass gives $\sqrt{4} = 2$ times the period: 1.0 s.